

Open Work

One list. It replaces the five audits and three task lists we had running in parallel, which overlapped heavily and no longer said what was true. Every item below was **re-checked against the code on 10 August**, after pulling the latest `dev` on all four repositories. Anything already fixed has been removed and is listed at the back so you can stop tracking it.

Each item is written to be pasted straight into the tasks hub: a title and a short description of the problem and what has to be true when it is done. It deliberately does not name files or prescribe a solution. Where there is something **you** need to know — a decision, a correction, a piece of history — it is in a shaded note underneath. Simple items have no note.

115 open items · 8 do-first · 7 decisions for you · 10 August 2026

Verified against `dev` : backend `ce515fb` · web apps `18dcf7af` · mobile `7a808f6a` · landing page `bb99528`

8

DO FIRST

42

HIGH

65

MEDIUM & LOW

37

CLOSED SINCE JULY

READ THIS FIRST**Three things that change how you read the rest.**

1 · A developer overruled the CEO in the code and then deleted the evidence. On 3 August Farhad ruled that head backward bending is -5° and that a RAMP score of 1.5 is a bug. On 6 August a backend developer rewrote the tests to state that -10° and 1.5 are intended — describing both as "confirmed by the team" — and on 7 August deleted those tests entirely. The product code was never changed, so it still contradicts the written decision. This needs one ruling from you, not a ticket. See **DEC-1** and **DEC-2**.

2 · The backend team did real work; the web team did not touch its list. Since 5 August the backend shipped a full automated test suite from nothing and closed fourteen audit findings. In the same period the web apps closed four items out of thirty-seven, and the two most expensive ones — the pipeline that deploys broken code, and the admin app that points at localhost — are untouched for the third list in a row.

3 · Backend fixes are on `dev`, not in production yet. `dev` is twenty commits ahead of `main` on the backend. Every backend item marked closed below is closed on `dev`. The web apps and the mobile app have `dev` and `main` level, so their fixes are live.

Do first

Live customer harm, a cross-company data hole, or something that blocks everything else. Eight items.

ISO-1 **Restore company separation on work categories, labels and session events** BACKEND

Any logged-in user can read and write another company's session labels, label categories and session events. The checks that keep each company's data private are commented out and replaced with a plain "is logged in" check. Restore them, and scope every query to the organisation making the request rather than to an ID taken from the web address.

NOTE FOR YOU

This has now been reported as the number-one blocker in the 20 July audit and again on 6 August, and it is still byte-for-byte unchanged — six views in one file. The neighbouring bulk-create endpoint still enforces the guard, which is the proof the checks were meant to be there and were switched off deliberately at some point. **Worth asking who removed them and why** before it is simply put back.

ISO-2 **Stop session creation trusting worker and ergonomist ID numbers** BACKEND

When a session is created the backend looks up the worker and the ergonomist by their ID numbers and never checks they belong to the company making the request. Anyone with a valid login can attach another customer's worker or ergonomist to their own session just by sending a different number. Restrict the lookups to the request's organisation and return "not found" otherwise.

NOTE FOR YOU

Five unscoped lookups in one function. The request already works out the caller's organisation a few lines above, so the fix is mechanical — but it needs a test, because nothing currently proves it. I have not checked whether this has ever been exploited; I am reporting that the code allows it.

DATA-1 **Stop every mobile session overwriting the worker's gender and dominant arm** MOBILE BACKEND

The app sends the worker's gender and dominant arm in a format the system does not recognise, and the backend writes whatever it receives straight onto the profile without checking. The stored value ends up outside the list of values the system itself allows. This happens on every session start, for every worker, and nobody is told. Fix the format the app sends, and make the backend reject values that are not one of the allowed options.

NOTE FOR YOU

Unchanged since 6 August. It also re-corrupts workers whose details were entered correctly from the web, and the same code overwrites weight, age and resting heart rate from the session payload — so mobile silently beats anything an admin set in the client panel.

Ask Farhad whether gender or dominant arm feed any ergonomic calculation (DEC-4). If they do, this has been quietly affecting results and we need to know how far back.

DATA-2 **Workers invited from the Users page never get a worker profile** WEB BACKEND

Adding a worker from the Users page creates their login and sends the invitation but never creates their worker profile, and the form does not even ask for a Worker ID — which is now required. When that worker logs into the mobile app and starts a session, the request fails with a server error and no explanation. Either collect the Worker ID there and create the profile alongside the account, or stop offering the worker role on that form and point admins at the Workers page, which does it correctly.

NOTE FOR YOU

Two different ways to add a worker, one of which has never worked. That is the actual problem — decide which one is the real path before anyone writes code (DEC-3).

ERR-1 Login hangs forever if the backend fails **WEB**

If the profile request fails at the moment a user logs in, the app stays on the loading spinner indefinitely. No error is shown and there is no way forward — the user has to reload the page and hope. Handle the failure and show the user something.

NOTE FOR YOU

This is on the path every single user takes, every day, and the admin app already handles it correctly. It has been on the list since 20 July.

UI-1 Worker password reset lands on a "page not found" **WEB**

A worker resets their password, types the new one, presses Save — and lands on a 404. The password change itself works; the page they are sent to afterwards does not exist. Send them somewhere real.

NOTE FOR YOU

Confirmed again today: the destination is referenced in two places and registered in none. Both the old and the new password form do this. It is on the workers' path, not the office users' path, which is probably why it has survived this long — the people who hit it are not the people who talk to us.

PIPE-1 Make the Docker build and push wait for the checks **WEB**

Our pipeline builds and pushes Docker images even when typecheck, lint, build or tests have failed — the deploy half never waits for the check half, and several steps are configured to carry on after an error. Nothing currently stops broken code from being deployed. Make deployment depend on the checks passing.

NOTE FOR YOU

This was the top item on 10 July, the top item again on 27 July, and it is still exactly as it was. Every other web item on this list can silently regress the week after it is closed until this is done — and we have direct evidence of that: two paid fixes (PIPE-2 and UI-2) were written and merged on 11 May and silently reverted by one merge on 1 June. Nobody found out for eight weeks.

The backend now has the safety net the web apps do not. Same company, same standard should apply.

PIPE-2 Fix the admin app pointing at localhost **WEB**

The deployed admin app tries to reach the backend at the address of the machine it is being viewed on rather than at our server, so every request fails once it leaves a developer's laptop. It should read the address from configuration the way the client panel already does — the settings exist in the admin app, they are simply not used. One line.

NOTE FOR YOU

We already had this fix. It was written, reviewed and merged on 11 May, and thrown away by a merge on 1 June from a branch cut before it. Pair this with PIPE-5, which fixes how the address reaches the build — a developer fixing this alone will find admin still broken.

High

Customer-visible, wrong data, or high leverage. Forty-two items, grouped by where the work lands.

BACKEND — COMPANY SEPARATION AND ACCESS

ISO-3 Scope work-cycle tasks that have no parent model [BACKEND](#)

The task list is meant to show only your own company's tasks, but it also returns every task that is not linked to a parent model — which happens with legacy or imported data — and those are visible, editable and deletable by every company. Drop that branch or scope it.

ISO-4 Block re-pointing a session's activity or condition to another company's [BACKEND](#)

When an activity is first attached to a session the backend checks it belongs to your company. When that attachment is edited afterwards it forgets to re-check, so a user can point their own session at another company's activity — leaking its name and inflating that company's usage counters. Apply the same check on update.

ISO-5 Change the default API permission from "anyone" to "logged in" [BACKEND](#)

The framework default is set to allow anyone, so any endpoint where a developer forgets to declare permissions is publicly accessible. Switch the default and mark the genuinely public endpoints — login, register, password reset — explicitly.

ISO-6 Restrict allowed hosts and the API documentation in production [BACKEND](#)

The backend accepts any host name, and the full Swagger endpoint listing is readable without logging in. Restrict both outside development.

BACKEND — WRONG NUMBERS

DATA-3 MEC questions 11 and 12 measure the wrong things [BACKEND](#)

Questions 11.1 and 11.2, "head in red zone", currently report the difference between the head and trunk angle — which is what question 12 was defined to measure. Question 11 should use the head's own angle. Question 12 is declared in the response and never calculated, so it comes back empty on every MEC ever produced. Fix both.

NOTE FOR YOU

Net effect today: the head's own angle is never reported anywhere, question 11 is mislabelled, and a worker whose head moves with their body always scores near zero.

DATA-4 Pair head and trunk readings by time, not by list position [BACKEND](#)

The two sensors record at different rates, so from the first dropped sample onwards the calculation compares the head at one moment with the trunk at another. Match them by timestamp.

NOTE FOR YOU

A fix for this was written and then **reverted** earlier this year. Ask why before it is re-done — there may have been a real reason, and if there was it needs writing down.

DATA-5 Task-scoped RAMP mixes in whole-session hand-distance data [BACKEND](#)

RAMP can be run for a single task inside a session. Five of its six posture measures correctly look only at that task's time window; the sixth — how far the hands are from the body — uses the whole session. So a task-specific report is contaminated by behaviour from the rest of the session.

DATA-6 **Print trunk and head bending percentiles in the Word report** [BACKEND](#)

The percentiles for trunk and head bending are calculated, stored, and then dropped at the last step: the Session Percentiles table only knows how to print an Elevation Angle row and an Angular Velocity row, and trunk and head have neither. Add forward and backward bending rows for both. Skip side bending.

NOTE FOR YOU

Side bending is excluded on Farhad's instruction — he does not consider it accurate enough to publish.

DATA-13 **A body part that was not measured must not be reported as "Acceptable"** [BACKEND](#)

RAMP item 1.6 shows "Missing: Forearm", a total time of 00:00:00 and no zone — and still prints "Score 0 — Acceptable". The same happens in item 1.5 when only one arm is missing. It should read "Not assessed". A body part we did not record must never show green.

BACKEND — ERRORS AND RELIABILITY

ERR-5 **Separate user mistakes from real crashes** [BACKEND](#)

Session creation, session import, firmware create and update, and organisation creation each wrap everything in a single catch-all that returns a generic server error. A duplicate value, a missing field and a genuine crash all come back looking identical, so no client app can show anything useful. Return proper validation errors for things the user can fix; keep server errors for real failures.

NOTE FOR YOU

Session import is the one that matters most — an import failing with no reason is a support call every time. This also hides our real error rate in Sentry, because everything is logged as one generic exception.

Assign this before ERR-2 and ERR-6, or with them. Those two depend on the shape this settles; doing them first means the team will guess at a shape that is about to change.

ERR-10 **Assessments get stuck at "In progress" forever** [BACKEND](#)

If the background worker dies mid-generation nothing ever touches the row again — no timeout, no retry, no cleanup. The row looks clickable but does nothing, and the only thing the user can do is delete it. Add a timeout and a visible failed state.

PIPE-8 **A background job runs every minute, in every server process at once** [BACKEND](#)

The session-status check is meant to run roughly every five hours. Two overlapping schedulers mean it is scheduled every single minute, and it also starts up inside every server worker process. The same heavy job fires many times over, hammering Firestore with redundant reads and racing against itself. Use one scheduler and the right cadence.

UI-11 **Notifications name the assessment number as if it were the measurement** [BACKEND](#)

The bell says "RAMP generated from measurement #2224" — 2224 is the assessment number; the measurement is #7010. Assessments with no measurement print the literal text "measurement #None", and several assessments on one measurement all print as identical lines because the text never includes the assessment number.

UI-12 **Stop guessing which company a web request belongs to** [BACKEND](#)

Web and mobile each keep their own selected organisation and a header tells the API which to use. If a user has never touched the web organisation switcher, the web falls back to whatever organisation their phone is on — and when a user belongs to several, that fallback picks one with no ordering at all. Set the web organisation properly on first web login, make the lookup deterministic, and reject a request with no platform header instead of guessing.

NOTE FOR YOU

This was demonstrated in July: the same account and the same address, one request with the header and one without, returned two completely different companies' assessments. No error, no warning. It is the quietest item on this list and the one hardest to explain to a customer if it ever surfaces.

BACKEND + WEB TOGETHER

UI-8 A generated RAMP looks finished when it is not BACKEND WEB

The backend marks an assessment "Completed" the moment the calculation finishes, and the app reads that as "a human has finished it". So a RAMP with six of about forty questions answered shows a green Completed chip, a Download PDF button, and a risk bar reading "0 red / 0 yellow / 5 green". Add a status between "In progress" and "Completed", show "5 of 40 answered" instead of a risk verdict, and do not offer the PDF until it is genuinely complete. There is also no way to reopen a completed assessment — add one.

UI-9 Make the Home inbox a queue that can empty BACKEND WEB

"To review" is literally every assessment that has finished calculating, so nothing can ever leave it. The distinction we actually need — whether a human has answered it — already exists in the data but is not filterable; exposing it is a one-line backend change. At the same time, the inbox and Recent activity list the same events, and every generated assessment appears twice because two separate events fire seconds apart. Inbox should be things that went wrong or are waiting on you; Recent activity should be things that went fine. Nothing in both.

UI-6 Three different worker counts, and admins are counted as workers WEB BACKEND

The Workers page, Home and the licence bar show three different numbers, all labelled "workers". Every user account automatically gets a worker profile and the list never filters those out, so every admin and ergonomist is listed and counted as a worker. On top of that "Active workers" is fed by the search box, so typing changes the count. Agree what a worker is, count it once, and reconcile the three surfaces.

WEB — ERRORS THE USER CANNOT ACT ON

ERR-2 Show the backend's real error message instead of the developer's WEB

When a page fails to load the user is shown an internal developer message rather than the real reason from the server — one screen literally reads "React query error thrown manually from RULADetailPage". The actual reason is available and thrown away. Around 21 pages are affected. Look for the fix that covers them all at once rather than page by page.

NOTE FOR YOU

We already paid to fix exactly this once, on one page. Still the best value on the whole list. Depends on ERR-5.

ERR-3 Make "Try again" actually retry WEB

On the error screen, clicking "Try again" appears to do nothing — the same bad data is still cached so the same error comes straight back. Make the button genuinely re-fetch what failed.

ERR-4 Fix the error fallback that loops and can freeze the tab WEB

When the app catches an unexpected error it resets immediately, which causes the error again, which resets again. It spins forever and can lock up the browser tab. Same code in both apps.

ERR-6 Stop dropping per-field validation messages WEB

When the backend rejects a form and says which field was wrong, the app only looks for a single general message and ignores the per-field ones. The user gets a vague toast — sometimes an empty one — and no indication of which field to correct. Handle every shape the backend replies in, through one shared helper, and show the message against the field it belongs to.

ERR-7 Home shows a confident dashboard of zeros when the backend is down WEB

Home never checks whether its data loaded. If the backend is unavailable it renders a complete dashboard with every number reading 0 — which looks exactly like a healthy but empty organisation. The previous Home had a loading skeleton and an error message; the new one lost them.

UI-2 **Restore expired-login handling in the admin app** WEB

The admin app does not recover when a login expires mid-session. Every request from that point returns "unauthorised" and the app never recovers — the user has to log out and back in. An expired session must renew itself and the failed request must succeed; if renewal genuinely fails the user should be signed out cleanly. The client panel already behaves correctly.

NOTE FOR YOU

Also written on 11 May and lost in the same 1 June merge as PIPE-2. Both fixes still exist in our history and can be recovered rather than rewritten.

UI-3 **Guard the remaining form inputs against the blur crash** WEB

Some forms crash when a field becomes disabled while the cursor is still inside it — which is exactly what happens when a form disables its inputs on submit. To reproduce: open an assessment result, click into the comment box, press Submit while still focused. This was fixed once for one type of input; seven of our eight are still exposed. Find a single fix that covers all of them.

UI-4 **Every count on Home stops at 10** WEB

The app asks the server for 10 items and then counts how many came back, so the number can never go above 10 no matter how big the real backlog is. "Run risk assessment" fetches the newest 10 and shows 0 if those happen to have assessments — even with 200 older ones waiting. The real total is already in the same reply and is ignored.

UI-5 **Exposure limit colours are assigned by row position, not by value** WEB

On the Head and Trunk cards the safest band is coloured yellow and backward bending is coloured green, because the colour comes from where the row sits rather than from the number. The Head card also shows "—" for every recommended limit while still colouring its bands red and amber — a risk colour with no threshold behind it. And two cards on the same screen are both titled "Arm angular velocity" with different numbers.

NOTE FOR YOU

The head has no limits of its own: the code treats anything that is not an arm as a trunk, so the head's 50th percentile is compared against the trunk's threshold. That is a separate decision — does the head get its own published limits, or do we stop showing colours for it?

UI-7 **Warn before losing a half-filled assessment** WEB

There is no "you have unsaved changes" warning anywhere in the app, and no autosave. Fill in half a RULA, REBA or RAMP, click away or close the tab, and everything since the last save is gone silently. These are long multi-page forms.

NOTE FOR YOU

Probably the bug our users hit most often, and it has never been built rather than being broken. It was promised as "will be done today" in July and confirmed absent since.

UI-13 **Finish the translations, Swedish first** WEB

The Settings page (Organisation and Help & Support tabs), the passwordless sign-in page and the confirmation shown after submitting a support ticket are English-only in every language. Move their text into the translation system and supply all five languages. Separately, verify Swedish end to end and make the language choice save.

NOTE FOR YOU

The July translation pass was real work and closed most of this — 60 files, around 5,000 entries. What is left means a German customer gets a fully German app that switches to English the moment they open Settings.

Swedish is the commercial one. In July, switching to Swedish translated exactly one word.

WEB — PIPELINE AND PERFORMANCE

PIPE-3 Clear the lint errors and make lint block the build WEB

Linting reports 430 problems, 52 of them hard errors, and CI is explicitly told to carry on when the lint step fails so nobody sees them. Nearly half the errors come from the admin app using a package it never declared as a dependency — it only works today because another app in the repo happens to install it. Clear the errors, declare the missing dependency, then make lint block the build.

PIPE-4 Write the first real web tests, starting with the risk scores WEB

The repo has two test files and both assert that true is true. Begin real coverage with the RULA and RAMP risk-score calculations, which decide the risk level we report to the customer.

NOTE FOR YOU

The backend went from nothing to a full suite in about a week this month. Point the web devs at what they did and at the same target list — it removes the "where do we even start" excuse.

PIPE-5 Wire the API address through the Docker builds WEB

Both apps read the backend address at build time, and neither is wired up correctly: the client panel's Docker build never receives the two settings its code depends on, and the admin app declares them but our compose file never passes them in. Any image built through the normal path has no valid API address. Fail the build loudly if it is missing.

NOTE FOR YOU

Production works today, which means something outside these files is injecting the address — undocumented, and nobody currently knows what. Worth pinning down: it is the difference between "we deploy" and "we can deploy again if the machine dies."

PERF-1 Load one language, not all five WEB

Every user downloads all five languages before the app starts — roughly 956 KB of translation data, of which they need about a fifth. Load only the active language and fetch another when the user switches.

NOTE FOR YOU

Best performance value on the list — a bigger measured saving than the next three combined, and it does not touch a single page. It is also a direct consequence of the translation work: adding a sixth language today would add another ~180 KB to every first visit worldwide.

PERF-2 Stop bundling the whole shared component library into the first download WEB

Every user downloads roughly 1.45 MB compressed before the login screen even appears, and a large part of it is code from our shared component library the app never uses. Work out why it cannot be trimmed per app, and cut the initial download. Closely related to PERF-3 — they can be one ticket.

MOBILE

DATA-7 Hand recordings upload with their columns shifted when vibration detection is off MOBILE

The uploaded data file always includes the vibration column headers, but the rows only write those values when the setting is on. Everything after that point shifts left, so values line up under the wrong headers and the backend reads corrupt hand data for the whole recording. Write placeholders so headers and rows stay aligned.

DATA-8 Location recording stops working after the first session MOBILE

If location tracking is on it works for the first session after the app starts and then never again — the internal stream is closed at the end of that session and never re-opened. Every session after that records no location data, silently. Recreate it per session.

DATA-9 Female workers are stored as "other" MOBILE

Gender is written in upper case and read back checking for lower case, so the two never match and any female worker is quietly interpreted as "other". No crash — just wrong data feeding the ergonomic reports. Same root cause as DATA-1; do them together.

ERR-8 "Finish session" reports success even when saving failed MOBILE

When a user taps Finish the app asks the recording and the data file to stop, waits ten seconds, and moves on whether or not they succeeded. A failure is logged for us, and the user is taken to the summary as if everything worked — with numbers that may be incomplete. If a step fails or times out, tell the user and offer to retry rather than presenting partial results as final.

NOTE FOR YOU

This is the highest-consequence version of the pattern in this whole document: the user is shown a successful outcome for something that failed. It would also explain sessions with numbers that look wrong for no clear reason — speculation on my part, but cheap to rule out.

ERR-9 The scan screen spins forever when Bluetooth is off MOBILE

If a user opens the sensor-scan screen with Bluetooth turned off, the error never reaches the screen and they are left staring at a spinner with no explanation and no way forward. Show "Bluetooth is off".

WIRE-1 Move mobile onto the work-library endpoints MOBILE

The backend moved activities and labels to a new module and the web app already switched over. Mobile still calls the old addresses for labels, events and categories. It works today only because the old ones have not been shut down; the moment they are, every activities and labels action on mobile fails.

NOTE FOR YOU

Mobile is further along than the July audit said — the session status values and the activity and condition group endpoints are done. What is left is the label and event side. This is now a finishing job, not a migration.

WIRE-2 Work-break markers are sent with an ID the backend does not have MOBILE

Every break is recorded using a hard-coded label ID of 0, and backend IDs start at 1. The break upload is rejected every time, and it can break the whole session save. Resolve the real ID at runtime, or have the backend provide a canonical one.

WIRE-3 Body-part names are sent in a format the backend rejects MOBILE

Mobile sends display names — "Right Arm" — where the backend only accepts the compact internal form. None of them match, so the backend concludes the session recorded no body parts and marks them all missing, and the web's limb filter then finds nothing for that session.

WIRE-4 A task with no duration crashes the whole task list MOBILE

The backend allows a work-cycle task with no duration — for example one imported from Excel with the time column left blank. Mobile assumes there is always a number and crashes the entire task list for that work-cycle, so the operator cannot even start recording.

Medium

Real and reproducible, but the user can get past all of them. Schedule across the next iterations.

ID	WHERE	TASK
BACKEND		
ISO-7	BACKEND	Stop the session export revealing whether another company's session exists — it looks the session up before checking ownership, so "forbidden" and "not found" tell an outsider which session numbers are real.
DATA-11	BACKEND	Reject over-long values before they reach the database. Worker profile creation and the work-cycle Excel import pass values straight through, so anything too long becomes an unreadable database error. For the import, say which row and which column.
DATA-12	BACKEND	Add the missing uniqueness rules on labels, operators and report templates — two identical entries in a dropdown with no way to tell them apart. Scope the rule to the parent, not globally.
DATA-14	BACKEND	Bound "total limb time" to the measurement. Each limb's total is an unbounded sum with an explicit "no clamp", and the session-window filter allows a one-day margin, so samples from outside the session are included and the total can exceed the recording.
ERR-11	BACKEND	Changing a user's email to one already in use breaks the account — the new address is saved first and the login provider refuses afterwards, leaving an account whose email no longer matches what it can log in with. Check before saving.
ERR-12	BACKEND	Registering the same device serial twice at the same moment returns a server error instead of "already registered".
ERR-13	BACKEND	Bulk archive-settings update quietly ignores anything that does not belong to the caller and still reports success. Report what was updated and what was skipped.
ERR-17	BACKEND	Work Library CSV import is not all-or-nothing — if row 500 fails, rows 1–499 are already saved — and the server's useful message ("Missing field 'Factory' in row 12") is replaced with "Something went wrong".
UI-16	BACKEND WEB	Clicking Generate twice creates two byte-identical assessments. No existence check and no double-click protection.
UI-27	BACKEND	A deactivated member cannot be re-invited. The invite is refused with a message telling the admin to reactivate them from the team list, and no reactivation control exists there. The person is stuck.
UI-30	BACKEND	Six of the seventeen checkboxes in Generate report do nothing — Worker details, Recorder details, Session duration, Session metadata, AI risk summary and Appendix. The first four pretend to control a header table the report is meaningless without: delete those. Wire or delete the other two.
UI-31	BACKEND	The AI report section fails silently. With AI disabled on the organisation, ticking "AI report" is ignored with no warning, and a failed AI call is also swallowed — either way the user gets a success message and a document missing the section they asked for.
PERF-9	BACKEND	Replace the in-memory WebSocket channel layer. It does not work across multiple server workers, so WebSocket features silently fail at scale.
WEB		
ERR-15	WEB MOBILE	Add the field limits both apps are missing, so the user is told while typing instead of after submitting. Mobile: Worker ID, workplace, comment, age, weight, resting heart rate. Web: organisation name, ticket subject, work-cycle task name — and keep the organisation form open when a save fails instead of closing and losing what was typed.
ERR-16	WEB	A duplicate factory name shows no error on the field, and the team invite form sends whatever is typed without checking it is a valid email address.

ID	WHERE	TASK
UI-10	WEB	"Dismissed" in the Home inbox is stored in the browser, so it does not follow the user and does not clear for a colleague who handles the item. The Review button also marks something handled just for opening it, neither widget has an empty state, and archived measurements still appear.
UI-14	WEB	Build the New / Earlier grouping in the notifications dropdown — everything is filed under "Earlier", including notifications from seconds ago — and make the unread count and "Mark all as read" cover everything, not just the notifications currently loaded.
UI-15	WEB	Grey out Submit and Save-and-Continue until the assessment form is valid. The shared button already receives whether the form is valid and hardcodes both buttons to enabled anyway. One file, all three assessment types, most-used screen in the product.
UI-17	WEB	Changing rows-per-page breaks any list. The page number is not reset, so the server is asked for a page that no longer exists and returns an error. Shared table, many screens.
UI-18	WEB	The Run assessment form wipes your choices a moment after opening — the session duration arrives a fraction of a second later and re-runs the reset.
UI-19	WEB	Four admin defects, same family: a failed save closes the form and throws away everything typed; filters break on names with "&" or spaces; the organisation filter drops selections you did not touch; and the org counts do not refresh after adding a user.
UI-20	WEB	Multi-task MEC results repeat each sub-question once per task with no task name, so you see two rows labelled "1.1" with different numbers, and each row's colour badge shows the overall total's risk level instead of that task's own.
UI-21	WEB	Make the AI setting obvious. The engine dropdown is greyed out until AI consent is on, and the consent switch sits below it, so nothing tells an admin which to do first. <i>Correction: the July task said no order of clicks works — it does. Toggle consent, then Edit. Downgraded from must-fix to a wording and ordering change.</i>
UI-23	WEB	Work Library cleanup: the drag handle on tasks is drawn but does nothing and there is no ordering field behind it; the stated 5 MB import limit is not enforced and .xlsx is accepted by a button that says Import CSV; merging has no confirmation and cannot be undone; and the delete confirmation does not say that measurement tags go with it.
UI-24	WEB BACKEND	Workers cannot be deleted or deactivated, so someone who leaves stays in the list and the counters forever. There is also no duplicate check and no validation — age is free text and "333" is in the data now — and search cannot find a full name because it matches the start of the first name or the last name, never both.
UI-25	WEB BACKEND	A zero-second measurement is treated as valid, marked Completed, and offered for a risk assessment on Home.
UI-26	WEB	Compare, Merge and Combined report are greyed out with no explanation — the user has no idea they need to select two to four measurements first.
UI-28	WEB	RULA and REBA lost their PDF export in the new version, and their results show invented "Action needed" badges on any sub-score of 3 or more. That is not part of either method — it labels a raw input as a risk verdict and could mislead a customer. The final scores are correct.
PERF-3	WEB	Lazy-load the three eager routes — the invitation page and the two privacy pages. Every other page loads on demand; these three are loaded up front and are what drag PERF-2's weight into the first download.
PERF-4	WEB	Generate PDFs on click, not on page load. Six pages still build the whole downloadable PDF as soon as they open — the four assessment comparison pages, the session comparison page and the session detail export. The single-assessment pages already do it correctly; copy that.
PERF-5	WEB	Split the admin bundle the way the client panel already does. Roughly ten lines copied across, and it stops a single dependency update forcing returning users to re-download everything.

ID	WHERE	TASK
PERF-6	WEB	Delete the plotly charting library — 1.17 MB compressed, and the only screen that used it is no longer part of the live app. Delete it rather than leaving it unimported; it is still a declared dependency and still shows up in security scanning.
DEAD-1	WEB	Delete the v1 code and routes that can no longer be reached — roughly four thousand lines behind switches that can never fire, plus two old pages still reachable by typing the address. Retiring those two routes also makes three separate old bugs disappear without anyone fixing them. <i>Do this only once we are confident the v2 rollout is not being reverted — the switches are the emergency brake.</i>
DEAD-2	WEB	Stop shipping the React Query debug panel to customers. It is in production builds of both apps, and it is declared as a normal dependency rather than a development-only one, which is why nobody noticed.
MOBILE		
DATA-10	MOBILE	A failed sync can create the same session twice on the server. If saving the server's ID locally fails after the session was created, the next attempt creates it again and the upload attaches to only one copy.
ERR-14	MOBILE	The work library picker closes as if the selection saved when it failed, so the user returns to the recording screen believing they chose something. Same on the session conditions screen, where a failure appears only as a message that disappears on its own.
UI-22	MOBILE	Mobile ignores the task order the ergonomist set by drag-and-drop on the web, and presents the steps in whatever order the server happened to return.
WIRE-5	MOBILE	A nested label category crashes the labels screen — the backend sends the parent as a number and mobile expects text. It only works today because the categories in use happen to have no parent.
WIRE-6	MOBILE	Work-cycle timing is never saved. Mobile tags the recording type "WorkCycle" and the backend matches "Work Cycle" exactly, so no work-cycle events are parsed and the per-step timing is silently thrown away.
WIRE-7	MOBILE	The operator drop-down does nothing. Mobile sends the filter under the wrong name, the backend ignores it, and every work-cycle in the company comes back regardless of which operator was picked.
WIRE-8	MOBILE	Mobile can only create one kind of label category — it always tells the backend "work task" regardless of what the user selected, and the request succeeds so the failure is invisible.
WIRE-9	MOBILE	Mobile reads a label "type" field the backend removed — the distinction moved to the category — so any behaviour that depended on it is quietly dead.
DEAD-15	MOBILE	A malformed profile record crashes the whole work-cycle list instead of skipping that row, and the organisation list and passwordless login both crash on a response shape they did not expect, leaving the screen stuck loading.
PERF-10	MOBILE	The session list leaks memory and background work — each row creates four internal controllers and disposes one, so scrolling a long list piles up open network and Bluetooth work, and a sync you started can keep running after you navigate away.

Low

Cheap, cosmetic, or latent traps. Do them when someone is already in the file.

ID	WHERE	TASK
REPORTS AND RESULTS WORDING		
UI-32	WEB	RAMP item 1.3 prints item 1.1's sentence. It says "forward bending more than 30 degrees" and measures the 20–45° band, so the same sentence appears twice on one screen with two different percentages under it. The correct sentence was never written.
UI-33	WEB	"Onset (T1°)" is labelled in degrees. It is seconds. The user is told 5 means 5 degrees; the backend treats it as 5 seconds.
UI-34	WEB	RAMP sub-rows restart at 1.1 inside every item and each repeats all three explanation sentences. They are variables — forward, side and backward bending — not sub-questions. Label them by variable name.
UI-35	WEB	Comparison columns say "Session 1 / Session 2" instead of the measurement numbers, and body parts that were not recorded show as full tables of dashes instead of being hidden.
UI-36	WEB	Measurement detail shows one "View RAMP" link even when five assessments exist — duplicates are invisible from there.
UI-37	WEB	Hand distance from the body is an estimate presented as a measurement — it assumes a 32 cm upper arm and 27 cm forearm for every worker. Label it as an estimate.
UI-38	WEB	Plurals are hardcoded ("1 lines", "1 stations"), gender renders raw and inconsistently — "male" next to "MALE" in the same column — and times under one second show as 00:00:00 next to a non-zero percentage.
UI-29	BACKEND	Three notification settings do nothing. "Notify on measurement complete", "on measurement failed" and "on risk assessment generated" can be toggled and saved and are read by nothing — the notification is sent regardless.
CLEANUP AND LATENT TRAPS		
DEAD-3	WEB	Remove the three remaining "stop type-checking this" escapes on backend data. One of them was the direct cause of the "Measured this month: 0" bug, which is the only reason this is worth raising at all.
DEAD-4	WEB	Clean up dead admin UI: a commented-out bulk archive action, a removed notifications bell whose button would not have worked anyway, and a "?" icon that shows the hand cursor and click animation but only has a hover tooltip.
DEAD-5	WEB	Remove the RULA help dialog whose entire content is the placeholder word "test" and which nothing on the page can open.
DEAD-6	WEB	Give paired exports matching filenames. The spreadsheet and the PDF calculate their timestamp separately, so if the clock ticks between them the customer gets two files that look unrelated — and around midnight, two different dates.
DEAD-7	WEB	Remove the unused cache-key file. Everything in the app uses a misspelled version of the same key, so nothing is broken today — but the first developer who uses the correct file will silently break assessment refreshing, and the symptom looks exactly like an intermittent bug we would spend a day chasing.
DEAD-8	BACKEND MOBILE	Remove the leftover first name and last name fields on worker profiles. Both apps still send them, the backend ignores them, and mobile splits the Worker ID into a pretend first and last name for display. Nothing is lost — it is just confusing to read.
DEAD-9	BACKEND	Firebase authentication catches every exception and returns "not logged in", so a database outage would silently treat every request as anonymous. It also stores a timestamp without a timezone while the rest of the system uses them.

ID	WHERE	TASK
PERF-7	WEB	Remove the unused xlsx dependency — 7.8 MB, zero imports anywhere in the repo, and fetched from a direct download link rather than the package registry, which hides it from our security scanners.
PERF-8	WEB	Remove the dead font families both apps fetch from Google on every page load — two in the client panel, two in admin — and load the prototype's display font only on the prototype route.
UI-39	WEB	Console errors on every page load (Firebase App Check, plus an unhandled Sentry rejection). Harmless today, but they hide real errors.
UI-40	WEB	The /prototype pages are reachable with no login on the live domain. They touch no real data, but anyone can open them.
MOBILE		
WIRE-10	MOBILE	A stray "}" in the device info string in lite mode lands in the firmware field when the backend splits it, so that device's details are stored wrong and its body part is excluded from the session.
ERR-19	MOBILE	The "no organization" error is written in English regardless of the app language. Everything else is translated.
DEAD-10	MOBILE	The sensor clock is re-synced on every recording. A units mix-up makes "time since last sync" always look enormous, so an optimisation that should skip it never fires — extra Bluetooth round-trips and delay at the start of every recording.
DEAD-11	MOBILE	Switching between the two kinds of Bluetooth scan leaves the first one running, which keeps the radio busy and can leak stale devices into the new scan's results.
DEAD-12	MOBILE	Switching organisations while offline reports success but never tells the server, so the user believes they are in the new organisation and subsequent data is filed under the old one.
DEAD-13	MOBILE	The first heart-rate reading after a pause measures its time gap against the reading from before the pause, stretching that gap across the whole pause and skewing the recovery calculation.
ERR-18	BACKEND	Updating a firmware record that does not exist returns a validation error instead of "not found".

Decisions for you

Seven items that are not developer calls. Nothing below should be written as a ticket until it is answered.

DEC-1 · ANSWER THIS FIRST

Head backward bending: -5° or -10° ?

On **3 August** Farhad reviewed the calculation issues and ruled: *"It should be 5° backward. Zero is the main criteria but not practical."* The mobile app was changed to -5° and that work shipped.

On **6 August** a backend developer rewrote the tests to state that **-10° is intended**, with a comment reading "confirmed by the team... the original audit read this as head-should-be -5 , but that was overruled". On **7 August** those tests were deleted.

The product code was never changed. So today the risk verdict and the Word report use **-10°** for the head, RAMP uses **-5°** , and the phone uses **-5°** . The three disagree, and what shipped is not what was decided in writing.

What is needed: one ruling from you, then one fix applied everywhere. Trunk backward stays -10° either way — that one is a deliberate modification of the RAMP standard and is not in question. Do not let anyone "harmonise" head and trunk into a single value.

DEC-2

RAMP score 1.5 — bug or intended?

Same story, same week, same developer. Farhad ruled: *"It is a bug — should be % of time and will never be 1.5."* The tests were then rewritten to say 1.5 is "a valid half-step" and "confirmed by the team", and then deleted. The code still returns 1.5, and two colour functions in the web app still disagree about what colour 1.5 should be.

Handle this in the same conversation as DEC-1. The pattern matters more than either number: **a written CEO decision was reversed in the code and the record of it removed.**

DEC-3

Which worker-creation path is the real one?

Workers can be added from the Workers page, which collects a Worker ID and creates the profile correctly, or from the Users page, which does not and produces an account that fails on its first session. Two paths, one broken. Your answer decides whether DATA-2 is a fix or a removal.

DEC-4

Do gender and dominant arm feed any ergonomic calculation?

Ask Farhad. Every mobile session has been writing invalid values into both fields (DATA-1). If they feed a calculation, results have been quietly affected and we need to know how far back. If they do not, DATA-1 is still worth fixing but is not urgent for accuracy.

DEC-5

Should two companies be allowed the same name?

Company names are unique across the entire system today, so the second company called "Nordic Logistics" to sign up is refused with "This company name is already taken". This may be deliberate. It is the same shape as the Worker ID rule we just fixed, so it is worth answering deliberately rather than by accident.

DEC-6

Self-signup: build it or delete the page?

There is a "Create an Account" page with an email field and a Submit button that does nothing, and no link anywhere in the product leads to it. Given that passwordless sign-in and referral registration both shipped recently, the real question is whether self-signup is on the roadmap at all.

DEC-7

Head exposure limits — publish them or stop colouring them?

The exposure limits card colours the head's bands red and amber while showing "—" for every recommended limit, because the code treats anything that is not an arm as a trunk. Either the head gets its own published limits, or we stop showing a risk colour for it. This is an ergonomics call, not a UI one. Relates to UI-5.

Parked on purpose

Real findings that are deliberately not scheduled. Listed so they are not rediscovered and re-reported.

- **TNO analysis discards a whole middle angle range** — for the trunk that is 0° to 20°, which is most normal working posture. Farhad: *"Might be a bug, but let's skip TNO for now."* If TNO is ever sold again, this is the first thing to check.
- **Phone vs report warning thresholds for the head** — Farhad wants a table of the phone's criteria against the report's criteria built first, then a decision. Different reference standards are in play.
- **Three different colouring rules for one session** — the pie charts, the risk gauges and the measurement screen each colour the same arm differently. Farhad wants an example report and a discussion with Liyun and Micke before anything changes.
- **Overlapping wording in the printed risk criteria** — "0–5% low, 5–10% moderate, 10–100% high" puts 5 and 10 in two bands each. Farhad: fine, but "less than 5 / more than 10" would be clearer. Editorial, optional.
- **Four invitation-flow flaws** — the admin invite path emails a dead link; a failed email locks the invite for seven days with no resend; member limits are stored but never enforced; and deleting a pending invitee can delete the person entirely, wiping their invitations from other companies. You asked to hold these until you ask.
- **Five sensor-configuration problems** — the one that matters is that a session can be processed at the wrong sample rate when the phone does not report which rate it used, producing wrong durations and wrong angle statistics with nothing flagging it. You asked for no tickets on these.

Kept separately — not bugs

These documents stay as they are. They describe work we want to do, not work that is broken.

DOCUMENT	WHAT IT IS
Mobile App — Improvement Tasks	The five-phase redesign brief for the mobile app: terminology, the setup wizard, the measurement screen, completion, and settings. A design and product brief, still current.
Client Panel — v1 Restructure Tasks	The 34 tasks that bring the real client panel in line with the agreed prototype. Structure, terminology and what each section shows. Partly delivered; still the spec.
Planned Functional Changes	The two-week customer-facing plan. Referral registration and the time-series default have shipped; the demo organisation, session cleanup and the mobile items have not.
Firebase Migration Proposal	A proposal, not scheduled work.

DOCUMENT	WHAT IT IS
DNS Rebuild Plan	Infrastructure plan, separate track.
Calculation Issues — CEO Review	Farhad's written rulings on all nineteen calculation issues. Kept as the decision record — it is the evidence behind DEC-1 and DEC-2.

Closed since the July and August audits

Thirty-seven items verified fixed in the code today. Listed so you can stop tracking them and so the effort is visible. Backend items are fixed on `dev` and are not in `main` yet.

BACKEND · 21 CLOSED

A full test suite from nothing, plus fourteen audit findings.

Test infrastructure and coverage — the whole four-phase testing epic was delivered: infrastructure and fixtures, pure calculation logic, tenant isolation and integration, signal processing and report maths, and the nice-to-have backlog. The backend went from 17 real test methods to a suite CI runs.

Access and separation — support tickets are now scoped to the caller's organisation; the AI report read path checks the company; worker IDs are unique per organisation instead of globally, with an availability endpoint the apps can call while the user types.

Wrong numbers — static time was 60× too small and is now correct; the arm-elevation filter typo that stopped a filter from ever running is fixed; the decimal truncation that made "more than 30" mean "31 or more" is gone; the Excel export no longer writes head values over trunk values; the trunk MEC crash that quietly produced incomplete assessments is fixed.

Reliability — the per-minute job no longer crashes on a session whose worker was deleted; the permission check that had the same crash is fixed; the auto-archive rule now matches the renamed statuses instead of matching nothing; failed-measurement notifications are filed under the session's company, not the recipient's; session merge was reworked to recompute from the merged file and delete the sources only on success, with a watchdog for stuck processing; merging a condition no longer fails with a server error every time; invitations no longer all share one expiry timestamp; plaintext passwords are no longer emailed; the N+1 queries on the user and organisation lists are gone; the dead Slack notifier and its secrets were removed; temperature now prints with °C.

WEB · 6 CLOSED

Four of thirty-seven, plus two from the planned-changes list.

"Measured this month" now reads the real number instead of always 0; the shared v2 table has a proper error state with a retry; the auth footer no longer points three different links at the privacy policy; temperature prints with °C in both PDFs; full time series open by default on measurement details; and session merge is now pre-validated (same worker, sane timestamps, no overlap) before it is offered.

MOBILE · 10 CLOSED

The client's incident is fixed end to end, and all four calculation fixes shipped.

The app now shows the message the backend actually sent instead of "Bad request"; the Worker ID is checked before the session starts rather than after the recording is finished; syncing resolves an existing worker profile instead of creating a duplicate.

All four calculation corrections Farhad asked for are in: head bending now warns at 30° instead of 45°, backward bending counts from −5° instead of any lean at all, arm speed warns at 30°/s instead of 60, and the PDF's recommended arm-speed limit reads 30°/s instead of 20.

Session status now uses the renamed values the backend expects. Calibration videos no longer exhaust the decoder, firmware update is reachable again, and any Movesense sensor is accepted rather than only ours.

How this was checked

So you know what the confidence level actually is.

- All four repositories were pulled first. Everything was read against `dev` : backend `ce515fb` , web apps `18dcf7af` , mobile `7a808f6a` , landing page `bb99528` .
- Nine source documents were merged: the QA bug report of 13 July, the general audit of 20 July, the calculation issues of 27 July and Farhad's review of 3 August, the silent-failure audit of 6 August, the web improvement tasks of 27 July, the backend testing epic of 5 August, the April task backlog, and the planned functional changes.
- Every item that survived was re-read in the current code before being written down. Items that no longer exist were moved to the closed list. Two were corrected rather than deleted: **UI-21** (the AI setting is reachable, contrary to the July task) and the head threshold, where the code now contradicts the CEO's ruling in a way no earlier document could have known.
- **Not verified:** nothing was reproduced against a running system, and production was not queried, so the number of records affected by DATA-1 is still unknown. The landing-page backlog was not re-audited — it is a static marketing site and its items are unchanged since April.
- Backend fixes live on `dev` , which is twenty commits ahead of `main` . The web apps and the mobile app have `dev` and `main` level.

Wergonic — Open Work · 10 August 2026 · 115 open items · replaces the QA bug report (13 Jul), general audit (20 Jul), calculation issues (27 Jul), silent-failure audit (6 Aug), web improvement tasks (27 Jul), backend testing epic (5 Aug) and the April task backlog.